

Gradiometer & Soil Analysis Project Database

This repository contains the structured technical data for the **Gradiometer & Soil Analysis System**, extracted from a comprehensive development chat log.

Structure of `project_database.json`

The database is organized into five main root keys:

1. **meta** : Contains project metadata, versioning, and executive notes regarding the data extraction process.
2. **Hardware_Analog** :
 - **afe_rx1** : Detailed specifications for the Analog Front End (AFE) receiving stage, including the AD8429 instrumentation amplifier and AD797 low-noise op-amp. Includes gain calculations and pin configurations.
 - **components** : A list of major ICs and modules (STM32H7, K210, RM3100, etc.) with their packages and functions.
 - **power_tree** : Details of the power management system, including isolated DC-DC converters and voltage regulators.
 - **connectors & pcb_layout** : Specifications for external interfaces and PCB design decisions.
3. **Firmware_Embedded** :
 - **mcu** : Primary and secondary microcontrollers used in the project.
 - **signal_processing** : Algorithms for gradiometry, cycle counting for RM3100, and excitation pulse timing.
 - **variables & protocols** : Critical code variables (like I2C addresses) and communication protocols used between modules.
4. **Android_App** :
 - Outlines the mobile application structure, including 3D visualization methods, data flow via BLE, and UI components.
5. **Decision_Log** :
 - A chronological record of technical pivots, covering component selection, power isolation strategies, and PCB workflow adjustments.

Usage

This JSON file is intended to be used as a "Source of Truth" for:

- Generating Bill of Materials (BOM).
- Configuring firmware constants.
- Documenting technical evolution for GitHub project management.

Technical Validation

The JSON file has been validated for syntax using:

Bash

```
python3 -c "import json; json.load(open('project_database.json'))"
```